

Rahul

Statistician — Medical Statistics — Real-World Data

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Profile

Medical Statistics and Health Data Science MSc graduate from the University of Bristol with experience analysing health and biomedical data using R. Skilled in regression modelling, survival analysis, epidemiological methods, model diagnostics and reproducible statistical workflows.

MSc dissertation used longitudinal ALSPAC/ARIES data to investigate maternal exposures and offspring DNA methylation. Seeking early-career Statistician opportunities in medical statistics, biostatistics and real-world data.

Education

University of Bristol, UK

MSc Medical Statistics & Health Data Science

2025–2026

Result: Pending

- Relevant modules: Regression Models; Statistics & Epidemiology; Applied Health Data Science; Advanced Statistical Methods; AI for Health Data Science; and Epidemic Data Analysis & Modelling.
- Training in study design, epidemiological methods, generalised linear models, causal diagrams (DAGs), model diagnostics, statistical programming and reproducible analysis.

Indian Institute of Technology Kanpur, India

MSc Statistics

2021–2024

Overall Grade: 60%

- Postgraduate training in statistical inference, probability, regression, generalised linear models, time series and computational statistics.
- Developed a strong theoretical foundation for applied statistical modelling and quantitative research.

University of Delhi, India

BSc (Hons) Statistics

2017–2020

CGPA: 7.095/10

- Studied probability, statistical inference, regression, sampling methodology, design of experiments and statistical computing.

Research & Statistical Projects

MSc Dissertation — Epigenome-Wide Association Study, ALSPAC/ARIES

University of Bristol

2026

- Investigated associations between maternal low-calorie/diet soft-drink consumption during pregnancy and offspring blood DNA methylation at birth, age 7 and adolescence.
- Worked with large-scale longitudinal epidemiological and epigenetic data from the ALSPAC/ARIES cohort.
- Conducted data quality checks, exposure and covariate preparation, regression modelling and adjustment for potential confounding factors.
- Applied multiple-testing considerations appropriate to high-dimensional epigenome-wide analyses and interpreted findings within an epidemiological framework.
- Used R to implement reproducible statistical analyses, produce diagnostic outputs and summarise results for scientific interpretation.

Breast Cancer Survival Modelling

- Analysed time-to-event data using Kaplan–Meier methods, log-rank tests and Cox proportional hazards regression.
- Investigated proportional-hazards assumptions and considered time-varying effects where appropriate.
- Compared conventional and machine-learning approaches to survival prediction, with emphasis on discrimination, calibration and clinical interpretability.
- Communicated results through reproducible statistical outputs and visualisations.

Statistical & Epidemiological Methods

Regression: Linear regression, logistic regression, Poisson regression, negative binomial regression, generalised linear models, interaction terms, confounding adjustment and model diagnostics.

Survival Analysis: Kaplan–Meier estimation, log-rank testing, Cox proportional hazards models, proportional-hazards assessment, time-varying effects and evaluation of survival predictions.

Epidemiology: Observational study design, measures of association, confounding, bias, effect modification, causal diagrams (DAGs) and interpretation of epidemiological evidence.

Additional Methods: Missing-data assessment, multiple testing, principal component analysis, clustering, predictive modelling, model validation and exploratory data analysis.

Professional Experience

Data Analyst Intern — Axtia – Ingenious Insights, India

May–Jul 2022

- Analysed and summarised large datasets to support business and data-driven decision-making.
- Used MicroStrategy to develop analytical outputs and dashboards for stakeholders.
- Supported data checking, reporting and documentation while working within a professional analytics team.
- Received a Pre-Placement Offer (PPO) following successful completion of the internship.

Teaching & Scientific Communication

Founder & Statistics Tutor — My Academic Tutor

2020–2025

- Taught statistics and quantitative methods including probability, statistical inference, regression and time series to learners from varied academic backgrounds.
- Developed structured teaching materials, statistical examples and R-based demonstrations.
- Translated technical statistical concepts into clear explanations, strengthening written and verbal communication skills relevant to multidisciplinary research.

Technical Skills

Statistical Programming: R (data manipulation, statistical modelling, visualisation, reproducible analysis); Python (intermediate)

Statistical Software: R, SPSS, Stata, SAS (introductory)

Data & Programming: SQL (foundational), Excel, MicroStrategy, Linux/Bash

Reproducibility: Git, GitHub, Quarto/R Markdown, Conda/Mamba, Snakemake

Research Skills: Statistical reporting, scientific interpretation, literature review, data quality assessment and reproducible research workflows